Hyperparameter Optimisation: A Comparative Study of Five Algorithms on YAHPO Gym

Hyperparameter Optimisation for Machine Learning — Assignment 1

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1 Introduction

We compare Random Search (RS), Grid Search (GS), Successive Halving (SH), Hyperband (HB), and Bayesian Optimisation (BO) on two YAHPO Gym surrogate benchmarks [4]: NB301 (neural architecture search, CIFAR-10) and rbv2_xgboost (XGBoost tuning, dataset 16). HB achieves the best final accuracy on NB301; on rbv2_xgboost the multi-fidelity methods converge faster without improving final quality.

2 Methods

Random Search (RS). RS [1] samples configurations uniformly at random and evaluates each at maximum fidelity, ignoring all prior observations.

Grid Search (GS). GS discretises each continuous dimension into 5 evenly-spaced values crossed with all categorical choices; NB301's 34-dimensional categorical space is handled by sampling each dimension independently.

Successive Halving (SH). SH [2] starts with $n{=}\lceil \eta^{s_{\max}+1}/(s_{\max}+1)\rceil$ configurations at $r_{\min}$, keeps the top $1/\eta$ fraction per round, and promotes survivors to $r \cdot \eta$ until the highest rung below $r_{\max}$ ($\eta{=}3$, $s_{\max}{=}\lfloor \log_\eta(r_{\max}/r_{\min})\rfloor$).

Hyperband (HB). HB [3] runs $s_{\max}{+}1$ SH brackets with $(n_s{=}\lceil (B/r_{\max}) \cdot \eta^s/(s{+}1)\rceil,\ r_s{=}\max(r_{\min}, r_{\max}{\cdot}\eta^{-s}))$, hedging against the sensitivity of any single bracket to its initial candidate count.

Bayesian Optimisation (BO). BO [5] fits a GP (Matérn-5/2 + WhiteKernel) to all past evaluations and selects the next configuration by maximising Expected Improvement over 1 000 random candidates (5 random warm-up evaluations; GP and EI implemented from scratch in NumPy).

3 Experimental Setup

NB301: fidelity = epochs ($r_{\min}{=}1$, $r_{\max}{=}98$), metric = val. accuracy. **rbv2_xgboost/16**: fidelity = training fraction, metric = accuracy. Budget $B{=}50{\times}r_{\max}$ fidelity units for all algorithms. Hyperparameter spaces accessed via ConfigSpace without the forbidden sampling methods; 10 independent random seeds per algorithm.

4 Results

NB301. All methods reach ≈93% within 500 budget units (Figure 1) but diverge afterwards. HB achieves the highest final accuracy (94.16±0.20); SH is weakest (92.79±0.17) because a single bracket exhausts its candidate pool and restarts with no memory, whereas HB always has other brackets running at different fidelity levels. BO matches HB in mean (94.08) but is less consistent (std = 0.32, Figure 2): NB301's 34 categorical hyperparameters produce a ≈272-dimensional one-hot vector that degrades the GP, making results highly seed-dependent. RS and GS perform nearly identically, as expected.

Table 1: Final incumbent (mean ± std, 10 seeds). Best in **bold**.

Algorithm	NB301	rbv2_xgb
Random Search	93.73 ± 0.17	0.962 ± 0.005
Grid Search	93.71 ± 0.18	0.963 ± 0.010
Succ. Halving	92.79 ± 0.17	0.968 ± 0.007
Hyperband	**94.16 ± 0.20**	0.967 ± 0.007
Bayesian Opt.	94.08 ± 0.32	**0.974 ± 0.009**

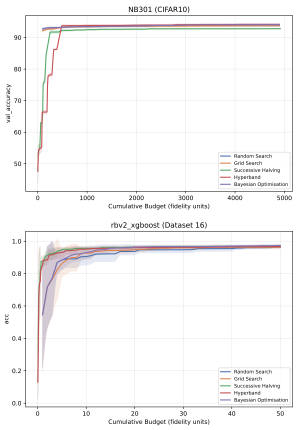

Figure 1: Incumbent curves (mean ± 1 std). Top: NB301. Bottom: rbv2.

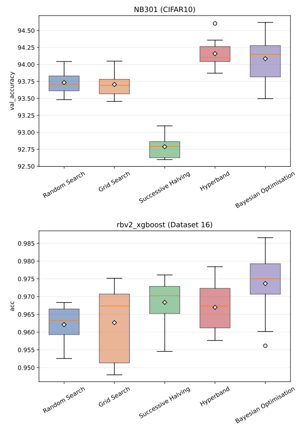

Figure 2: Final incumbent per seed. HB best on NB301; BO widest spread.

rbv2_xgboost. BO achieves the highest final accuracy (0.974±0.009), but the more interesting story is in convergence speed (Figure 1). SH and HB reach ≈0.96 after only 10 budget units, while RS, GS, and BO need 20–30 units to match that level. XGBoost rankings at small training fractions correlate well with full-data rankings, so multi-fidelity methods can cheaply discard poor configurations early and reach a strong incumbent fast.

5 Conclusions

HB achieves the best final accuracy on NB301; its bracket portfolio hedges against a poor initial candidate count, the key advantage over a single SH bracket. On rbv2_xgboost, BO reaches the highest final accuracy (0.974±0.009), but SH and HB find a good incumbent significantly faster, making them the practical choice when convergence speed matters. BO is unreliable on NB301 due to the ≈272-dimensional one-hot encoding degrading the GP surrogate, leading to high variance across seeds.

Statement of Contributions

Shivam Singh Gahlot implemented SH, HB, and BO, ran all experiments, and wrote the report. Dlyaver Seitveliev implemented RS and GS and contributed to presentation slides and visualisations.

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